Bilingual Recipe Translation Using Transformer-based Seq2Seq Modeling

W266
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Problem Statement

State of the art models today perform decent when it comes to translating text across multiple languages including recipes….

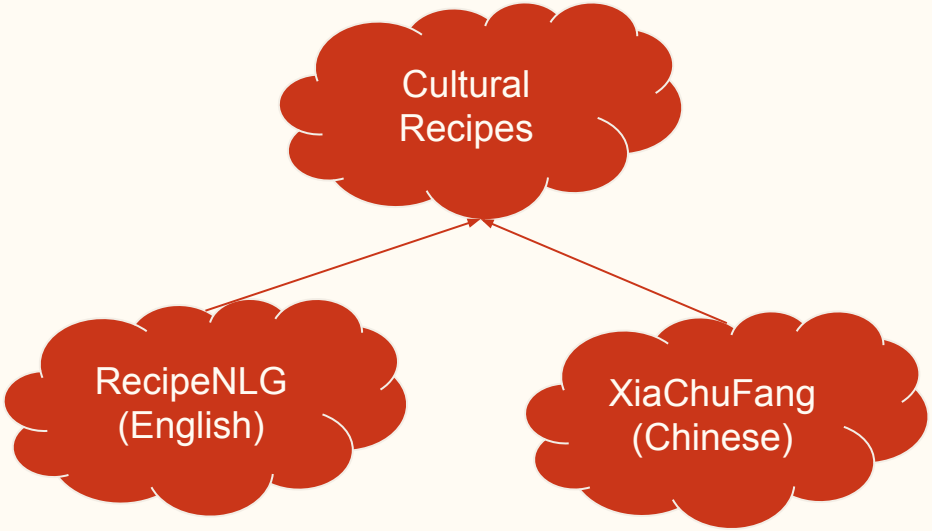
However, are they able to provide precise translations that adapt to regional cross-cultural differences?

Background

Cultural Adaptation of Recipies (Yong et. al, 2024)

- Chinese → English bidirectional translations
- Cultural Recipies Dataset
- We will enhance the modeling experiments by building up on existing architectures

Data



Data loading (JSON):
Flattening (1:1 pairs) → 265K total
- Dynamic padding tokenization
- 80/10/10 Training/Validation/Test split
- Similar pre-processing for each direction

Raw data divided into 3 sections:
1. Title
2. Ingredients
3. Steps

Human & Silver Evaluation sets
- Human: 25 cn2en, 41 en2cn
- Silver: 82 cn2en and 52 en2cn

*Note: Entire dataset was too large, so experimentation was performed on a subset of 2000

Methodology

Baseline:	Fine-Tuned:		
mT5-base	mT5-small	mT0-small	mBART-50
• Batch size of 32 • 30 epochs • Learning rate of 1e-4 • Adam optimizer	• Batch size of 32 • 3 epochs • Learning rate of 5e-4 • Adafactor optimizer	• Batch size of 32 • 2 epochs • Learning rate of 5e-5 • Adafactor optimizer	• Batch size of 32 • 3 epochs • Learning rate of 5e-4 • Adafactor optimizer

- ➔ Fair benchmarking structure
- ➔ Comparable analysis
- ➔ Fine-tuned from baseline model

Results

		mT5-baseline	mT5-small	mT0-small	mBART-50
human_cn2en	BLEU	1.60	4.19	4.94	4.30
	ROUGE1	25.26	30.39	28.68	28.23
	ROUGE2	4.29	6.41	4.97	5.41
	ROUGEL	17.84	22.39	20.25	20.70
human_en2cn	BLEU	15.2	11.68	1.80	7.07
	ROUGE1	32.97	35.66	28.55	20.35
	ROUGE2	11.93	13.43	12.37	3.11
	ROUGEL	31.69	33.49	27.73	16.68
silver_cn2en	BLEU	3.74	3.94	4.5	3.86
	ROUGE1	28.93	29.31	28.89	27.42
	ROUGE2	6.92	7.69	7.59	5.95
	ROUGEL	19.51	20.91	21.30	19.58
silver_en2cn	BLEU	16.71	11.19	9.01	6.72
	ROUGE1	48.31	47.92	45.87	12.23
	ROUGE2	22.71	18.15	22.09	0.53
	ROUGEL	45.72	44.69	45.16	11.37

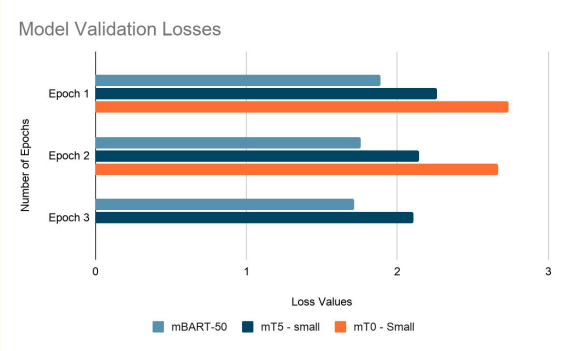
Our mT5-small outperformed the larger baseline on half the dataset!

Main Takeaways:

- Our mT5-small had highest scores overall
- mBART-50 performed the most poorly overall, even with the highest number of parameters

Decoding/Evaluation Strategy:

- Num_beams = 5
- No explicit target token-forcing or repetition_penalty



Baseline (mT5-base)

- Outperformed all models in silver_en2cn
- Potential overfitting in cn2en direction

mT5-small

- Best performing model throughout ***(notable: mT5-small has ½ parameters as mT5-base)***
- Fully trained for 3 epochs with a high LR (5e-4) for effective adaptation

mT0-small

- Best BLEU for cn2en
- Better metrics for the silver evaluation set over human

mBART-50

- Comparable results for cn2en, but worst for en2cn
- Near 0 ROUGE score for silver en2cn indicates EN → CN decoding issues

Analysis

Example Translation:

Input: 扬州炒饭, Ingredients: 半根胡萝卜 两个鸡蛋 半根黄瓜 小半个紫皮圆葱 半根秋林红肠, Steps: 紫皮圆葱、黄瓜、红肠切小丁, 胡萝卜切沫, 备用 鸡蛋打散, 用筷子高速搅打几下至出沫, 下油锅炒散, 盛出来, 备用。炒...
Output: Title: Turkish Fried Rice, Ingredients: 2 tablespoons olive oil, 1 medium onion, finely chopped, 3 garlic cloves, minced...

CLOSING *Thoughts*

mT5-small's success shows that more parameters does not indicate better performance, especially with insufficient or faulty training

BLEU and Rouge are both important to capturing the full story

Take the evaluation off-line! Taste test anyone?

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